KYEI BAFFOUR AFARI

Contact

Website: kbconsult2017.github.io
e-mail: kafari@unmc.edu
Phone: +1 (423) 440 4767
Department of Biostatistics
University of Nebraska Medical Center
984375 Nebraska Medical Center
Omaha, NE 68198-4375, USA

Miscellaneous

Citizenship: Ghana
Immigration Status: US Permanent Resident (in process)

Research Interests

Bayesian Statistics　Spatio-Temporal Modeling
Bayesian Kernel Machine Regression　Causal Inference
Environmental Health Statistics　Hierarchical Bayesian Modeling
Spatial Statistics　Dimension Reduction　Missing Data Methods

Education

University of Nebraska Medical Center

Ph.D. Biostatistics — 2026 (Expected)

Dissertation: Scalable Bayesian Spatio-Temporal Modeling

- Compressed Spatio-Temporal Basis Functions: An SVD-PICAR Framework for Scalable Gaussian and Multinomial Inference
- SVD-PICAR Bayesian Kernel Machine Regression Framework for Spatio-Temporal Count Data
- A Cluster-Based Framework for Joint Mixture Effects: Extending SVD-PICAR BKMR to Spatio-Temporal Count Outcomes

Relevant Coursework: Bayesian Theory and Methodology in Biostatistics, Categorical Data Analysis, Biostatistical Theory and Models for Survival Data, Advanced Biostatistics Theory, Biostatistics Computing (R Shiny, Python, and R), Theory of General Linear and Mixed Models in Biostatistics.

University of Cincinnati

M.Sc. Mathematics — 2023

Relevant Coursework: Advanced Calculus I & II (from *Principles of Mathematical Analysis* by Walter Rudin), Abstract Linear Algebra, Numerical Analysis, Mathematical Programming (Python), Partial Differential Equations and Fourier Analysis, Non-linear Optimization (Python), Design of Experiments, Linear Model Multivariate Analysis.

East Tennessee State University

M.Sc. Statistics and Mathematics — 2021

Thesis: Performance Comparison of Imputation Methods for Mixed Data Missing at Random with Small and Large Sample Data Set with Different Variability

Relevant Coursework: Mathematical Statistics, Survival Analysis (SAS and R), Statistical Methods I & II (R and Python), Real Analysis I & II, Theory of Matrices (Linear Algebra), Graph Theory, Fundamentals of Functional Analysis, Sampling and Survey Techniques.

University of Ghana

B.Sc. Mathematical Science (Statistics and Computer Science) — 2011

Relevant Coursework: Probability Distributions, Statistical Methods, Data Analysis, Algebra, Calculus, Computer Organization, Assembly Language Programming (Java).

Research Experience

UNMC Water, Climate and Health Program, ***University of Nebraska Medical Center*** 2023 – 2026

Graduate Research Assistant

- Conduct statistical analysis and methodological design for research projects using R, SAS, and SQL.
- Develop scalable Bayesian spatio-temporal methodologies for environmental and health applications.
- Design hierarchical Bayesian models using NIMBLE and advanced computational methods.

KB Statistical Consulting Ltd, Ghana 2017 – 2019

Statistical Research Consultant

- Managed, coded, and analyzed data using NVivo, STATA, and SPSS.
- Drafted methodologies and reports for statistical projects.

Statistical Department, ***NiBS Ghana*** 2016

Statistical Research Assistant

- Co-authored case studies with a team of statistical researchers.
- Drafted large portions of the case, interviewed, transcribed, and analyzed data using NVivo.
- Assisted approximately 45 doctoral students with article summaries and data analyses using SPSS, STATA, and NVivo.

Teaching Experience

University of Nebraska Medical Center

Graduate Teaching Assistant, Department of Biostatistics 2026 –

- Tutored graduate students in applying biostatistical methods to biomedical and public health research.

University of Cincinnati

Graduate Teaching Assistant, Department of Mathematical Sciences / Math and Science Support Center (MASS) 2021–2023

- Taught and graded a 3-credit-hour course in Quantitative Reasoning and Introductory Statistics.
- Tutored students at the Math and Science Support Center (MASS); conducted problem sessions and graded for a Calculus class.

East Tennessee State University

Graduate Teaching Assistant, Department of Mathematics and Statistics / Center for Academic Achievement (CFAA) 2020–2021

- Tutored students at the Center for Academic Achievement (CFAA) in Introductory Statistics and Linear Algebra.
- Assisted with Pre-calculus and Probability and Statistics (non-calculus) courses.

University of Ghana

Undergraduate Teaching Assistant, Department of Information Studies 2011–2012

- Tutored approximately 60 undergraduate students in programming and automation of information systems.
- Led a weekly training session for approximately 20 undergraduate students in database concepts and programming using Visual Basic .NET.

Refereed Academic Journal Articles

1. **Afari, K. B.** & Gwon, Y. (2026+). Compressed Spatio-Temporal Basis Functions: An SVD-PICAR Framework for Scalable Gaussian and Multinomial Inference. *Journal of Computational and Graphical Statistics* (Taylor & Francis). Under review.
2. **Afari, K. B.** & Gwon, Y. (2026+). SVD-PICAR Bayesian Kernel Machine Regression Framework for Spatio-Temporal Count Data. *Statistics in Medicine* (Wiley). Under review.
3. Gribben, K. C., Gwon, Y., **Afari, K.**, Berman, J., Li, A., Fard, B., Abadi, A.,

Wardlow, B., Tong, D., Tao, Z., & Bell, J. (2026+). Drought increases the risk of COPD mortality in the United States. *Environmental Research* (Elsevier). Under review.

4. **Afari, K. B.** & Lewis, C. N. H. (2022). Performance Comparison of Imputation Methods for Mixed Data Missing at Random with Small and Large Sample Dataset with Different Variability. *Asian Journal of Probability and Statistics*, 20(2), 16–39.
5. **Afari, K. B.** (2022). The role of fixed income in pension scheme investment in Ghana: A possible adoption for the United States economy. *Research Journal of Finance and Accounting*, 13(14), 27–35.

Talks & Conference Presentations

1. Contributed Poster Presenter. *American Statistical Association Joint Statistical Meetings 2026*, Boston, Massachusetts. "SVD-PICAR Bayesian Kernel Machine Regression Framework for Spatio-Temporal Count Data."
2. Contributed Talk. *ENAR 2026 Spring Meeting*, Indianapolis, Indiana. "Compressed Spatio-Temporal Basis Functions: An SVD-PICAR Framework for Scalable Gaussian and Multinomial Inference."
3. Oral and E-poster Presenter. *American Statistical Association Symposium on Data Science and Statistics 2025*, Salt Lake City, Utah. "Advancing Spatio-Temporal Modeling: A Hierarchical Gaussian Approach Using the SVD-PICAR."
4. Oral and E-poster Presenter. *Women in Statistics and Data Science Conference 2023*, Washington, DC. "Comparing the Performance of Imputation Techniques for Missing Data in Mixed-Type Variables Under the Missing at Random Assumption with Small and Large Sample Sizes."

Peer Referee

Journal of Applied Statistics (Taylor & Francis Group) 2023 –

- Reviewed manuscript: *A Hybrid Approach for Classification: Integrating Data Mining and Multi-Criteria Decision-Making Methods.*
- Reviewed manuscript: *Bayesian Treed Regressions for Estimating Heterogeneous Trajectories of Test Scores in Large-scale Educational Assessment.*

Mathematical Association of America (Adhoc Reviewer) 2021 –

- Reviewing the book: *Survival Analysis (Proportional and Non-Proportional Hazards Regression)* by John O'Quigley.

Honors & Awards

National Science Foundation (NSF) Travel Award — EAPOST Workshop on Teaching Statistical Thinking, Boston, Massachusetts 2026

Travel Award — American Statistical Association Symposium on Data Science and Statistics, Salt Lake City, Utah 2025

First-Year Departmental Summer Fellowship ($3,150), University of Cincinnati 2022

Certifications & Training

EAPOST Faculty Development Workshop — Teaching Statistical Thinking with American Statistical Association GAISE 2025

NASA ARSET — LIDAR Profiling Satellite Observations for Air Quality Application June 2025

NASA ARSET — Sea Level Changes Tools for Planning and Decision Support June 2025

NASA ARSET — Introduction to NASA Snow and Ice Data Products and Applications for Water Resource Management August 2025

STAT MOC Course on Satellites and Fire (American Meteorological Society) August 2025

MEMBERSHIPS

Sigma Xi, Full Member	2023 –
American Statistical Association	2023 –
Royal Statistical Society	2025 –
Biometric Society — Eastern North America Region	2025 –
American Mathematical Association	2021 –

ACADEMIC & COMMUNITY SERVICE

Member, Academic Statistics Consulting Group, ASA Statistical Consulting Section 2025 –

- Discuss topics of interest with academic consultants in a biweekly video conference.

Treasurer, UNMC College of Public Health (COPH) Student Association 2026

- Account for and distribute all COPH funds; manage student activity funds; propose the COPH SA budget; and sign, in conjunction with the President and ADSA, all disbursements.
- Attend and serve as a voting member of all Executive Board meetings.

Judge, Southwest Ohio Science and Engineering Expo, Ohio Academy of Science 2023

- Judged statistical projects in grades 5–8 and students in grades 8–12 to receive the American Statistical Association Cincinnati Chapter award for excellence in statistics in science fair projects.

Judge, Buckeye Science and Engineering Fair, Ohio Academy of Science 2023

- Assessed statistical projects for middle school students to receive American Statistical Association awards.

Volunteer, Daugherty Water for Food Global Institute (DWFI) Fall Welcome Event 2025

- Networked with faculty and students to share research work in support of DWFI's mission.

SOFTWARE SKILLS

R, Python, SAS, NIMBLE, STATA, SQL, SPSS, STATGRAPHICS, NVivo Java, C/C++, Visual Basic .NET

TOOLS

LaTeX, GitHub, Microsoft Excel